gravitational force field, the graviton, would be its messenger particle. At first sight, as we now note, this suggestion would appear to be particularly apt because the quantum field theory of the three nongravitational forces reveals that there is a tantalizing similarity between them and an aspect of the gravitational force we encountered in Chapter 3.

Recall that the gravitational force allows us to declare that all observers—regardless of their state of motion—are on absolutely equal footing. Even those whom we would normally think of as accelerating may claim to be at rest, since they can attribute the force they feel to their being immersed in a gravitational field. In this sense, gravity enforces the symmetry: it ensures the equal validity of all possible observational points of view, all possible frames of reference. The similarity with the strong, weak, and electromagnetic forces is that they too are all connected with enforcing symmetries, albeit ones that are significantly more abstract than the one associated with gravity.

To get a rough feel for these rather subtle symmetry principles, let's consider one important example. As we recorded in the table in endnote 1 of Chapter 1, each quark comes in three "colors" (fancifully called red, green, and blue, although these are merely labels and have no relation to color in the usual visual sense), which determine how it responds to the strong force in much the same way that its electric charge determines how it responds to the electromagnetic force. All the data that have been collected establish that there is a symmetry among the quarks in the sense that the interactions between any two like-colored quarks (red with red, green with green, or blue with blue) are all identical, and similarly, the interactions between any two unlike-colored quarks (red with green, green with blue, or blue with red) are also identical. In fact, the data support something even more striking. If the three colors—the three different strong charges—that a quark can carry were all shifted in a particular manner (roughly speaking, in our fanciful chromatic language, if red, green, and blue were shifted, for instance, to yellow, indigo, and violet), and even if the details of this shift were to change from moment to moment or from place to place, the interactions between the quarks would be, again, completely unchanged. For this reason, just as we say that a sphere exemplifies rotational symmetry because it looks the same